

because gamma decay do not change any of the above numbers.

- 30** Let  $n$  be the number of half-lives that would have passed when the 500 stockpile of bomb is reduced to 1:

$$\left(\frac{1}{500}\right) = \left(\frac{1}{2}\right)^n \Rightarrow n = 8.966$$

$$\text{Number of years} = \frac{(8.966)(1.7 \times 10^8)}{(365 \times 24 \times 60 \times 60)} = 48.3 \text{ year}$$